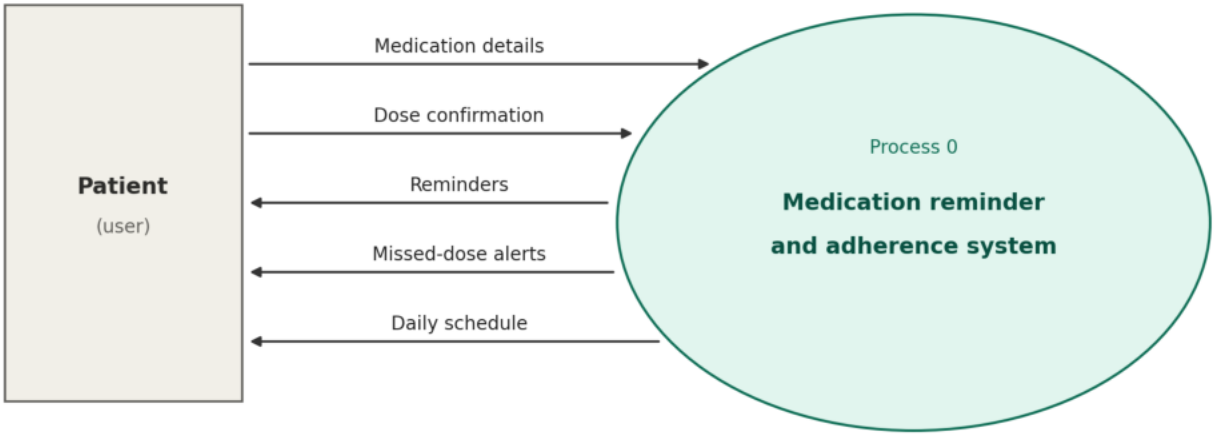
Context Diagram and DFD Level 0

Medication Reminder and Adherence System

1. Context Diagram

The Context Diagram shows the whole system as one process (Process 0) and the people or systems outside it that exchange data with it. In this system the only external entity is the **Patient (user)** .

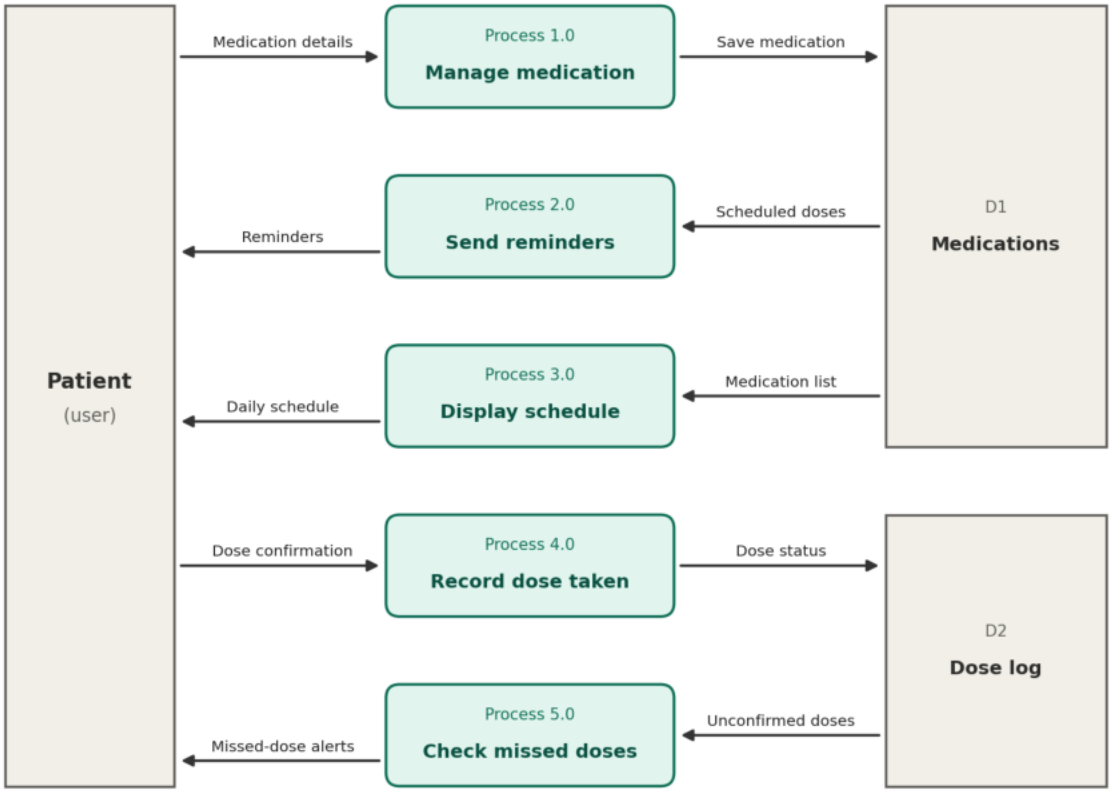


Data flows in the Context Diagram

Data flow	Direction	Description
Medication details	Patient to system	Name, dose and time of each medication (FR1)
Dose confirmation	Patient to system	The patient confirms that the dose was taken (FR3)
Reminders	System to patient	Reminder sent at the scheduled time (FR2)
Missed-dose alerts	System to patient	Repeat reminder when the dose is not confirmed (FR4)
Daily schedule	System to patient	List of today's medication doses (FR5)

2. DFD Level 0

The Level 0 DFD opens Process 0 into its five main processes and shows the two data stores that the processes use. The patient and the five data flows between the patient and the system are the same as in the Context Diagram, so the two diagrams are balanced.



Processes

No.	Process	What it does
1.0	Manage medication	Receives the medication details from the patient and saves them in D1 (FR1).
2.0	Send reminders	Reads the scheduled doses from D1 and sends a reminder to the patient at the scheduled time (FR2).
3.0	Display schedule	Reads the medication list from D1 and shows the daily schedule to the patient (FR5).
4.0	Record dose taken	Receives the dose confirmation from the patient and saves the dose status in D2 (FR3).
5.0	Check missed doses	Reads the unconfirmed doses from D2 and sends a missed-dose alert to the patient (FR4).

Data stores

ID	Data store	Contents
D1	Medications	Medication name, dose and scheduled time for each medication.
D2	Dose log	Status of each dose (taken or not taken) with date and time.

Data flows inside the system

Data flow	From	To
Medication details	Patient	1.0 Manage medication
Save medication	1.0 Manage medication	D1 Medications
Scheduled doses	D1 Medications	2.0 Send reminders
Reminders	2.0 Send reminders	Patient
Medication list	D1 Medications	3.0 Display schedule
Daily schedule	3.0 Display schedule	Patient
Dose confirmation	Patient	4.0 Record dose taken
Dose status	4.0 Record dose taken	D2 Dose log
Unconfirmed doses	D2 Dose log	5.0 Check missed doses
Missed-dose alerts	5.0 Check missed doses	Patient